

Cart (Classification & Regression Trees).

Day	outlook	temperature	humidity	wind	Decision
1	sunny	hot	high	weak	No
2	sunny	hot	high	strong	No
3	overcast	hot	high	weak	Yes
4	rainfall	mild	high	weak	Yes
5	rainfall	cool	normal	weak	Yes
6	rainfall	cool	normal	strong	No
7	overcast	cool	normal	wtrong	Yes
8	sunny	mild	high	weak	No
9	sunny	cool	normal	weak	Yes
10	rainfall	mild	normal	weak	Yes
11	sunny	mild	normal	strong	Yes
12	overcast	mild	high	strong	Yes
13	overcast	hot	normal	weak	Yes
14	rainfall	mild	high	strong	No

The dirtyness measures the homogeneity of a dataset.

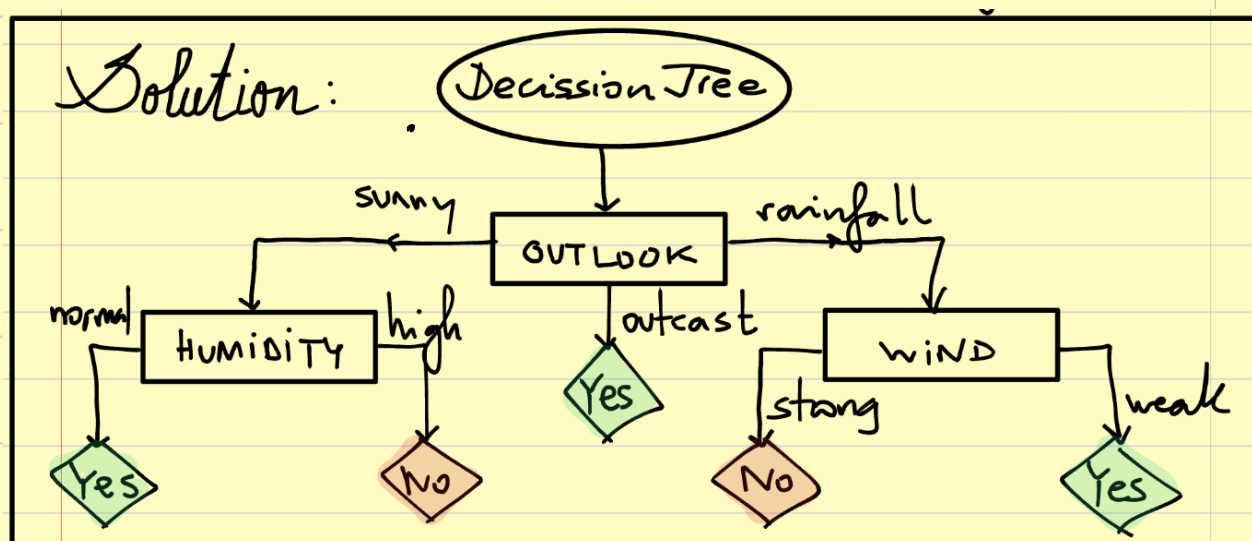
We measure it with the

Gini Index .
$$Gini = 1 - \sum_{i=1}^n p_i^2$$

$Gini = 0$ means the data is pure.

$Gini = 1$ means the data is dirty.

Goal: Decission tree.



Step 1. Look at all criteria and pick the one with the smallest Gini .

1. OUTLOOK. sunny, overcast, rainfall

OUTLOOK	YES	NO	#
sunny	2	3	5
overcast	4	0	4
rainfall	3	2	5
			14

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5	rainfall	cool	normal	weak	Yes
6	rainfall	cool	normal	strong	No
7	overcast	cool	normal	wtrong	Yes
8	sunny	mild	high	weak	No
9	sunny	cool	normal	weak	Yes
10	rainfall	mild	normal	weak	Yes
11	sunny	mild	normal	strong	Yes
12	overcast	mild	high	strong	Yes
13	overcast	hot	normal	weak	Yes
14	rainfall	mild	high	strong	No

$$\text{Gini}(\text{Outlook Sunny}) = 1 - \sum p_i^2 = 1 - \left[\left[\frac{2}{5} \right]^2 + \left[\frac{3}{5} \right]^2 \right] = 0.48$$

$$\text{Gini}(\text{Outlook Overcast}) = 1 - \sum p_i^2 = 1 - \left[\left[\frac{4}{4} \right]^2 + \left[\frac{0}{4} \right]^2 \right] = 0$$

$$\text{Gini}(\text{Outlook Rainfall}) = 1 - \sum p_i^2 = 1 - \left[\left[\frac{3}{5} \right]^2 + \left[\frac{2}{5} \right]^2 \right] = 0.48$$

$$\text{Gini}(\text{Outlook}) = \frac{5}{14} \cdot 0.48 + \frac{4}{14} \cdot 0 + \frac{5}{14} \cdot 0.48 = 0.342$$

2. Temperature . hot, mild, cool

TEMP	Yes	No	#
hot	2	2	4
mild	4	2	6
cool	3	1	4

$$\begin{aligned} \text{Gini}(\text{Temp}) = & \frac{4}{14} \cdot \left[1 - \left[\left[\frac{2}{4} \right]^2 + \left[\frac{2}{4} \right]^2 \right] \right] + \frac{6}{14} \cdot \left[1 - \left[\left[\frac{4}{6} \right]^2 + \left[\frac{2}{6} \right]^2 \right] \right] + \\ & + \frac{3}{14} \cdot \left[1 - \left[\left[\frac{3}{4} \right]^2 + \left[\frac{1}{4} \right]^2 \right] \right] = 0.439 \end{aligned}$$

3. Humidity. high/normal.

HUM.	YES	NO	#
high	3	4	7
normal	6	1	7

$$Gini(HUM.) = \frac{7}{14} \cdot \left[1 - \left[\left(\frac{3}{7} \right)^2 + \left(\frac{4}{7} \right)^2 \right] \right] + \frac{7}{14} \cdot \left[1 - \left[\left(\frac{6}{7} \right)^2 + \left(\frac{1}{7} \right)^2 \right] \right] =$$

$$= 0.367$$

4. Wind. weak/strong.

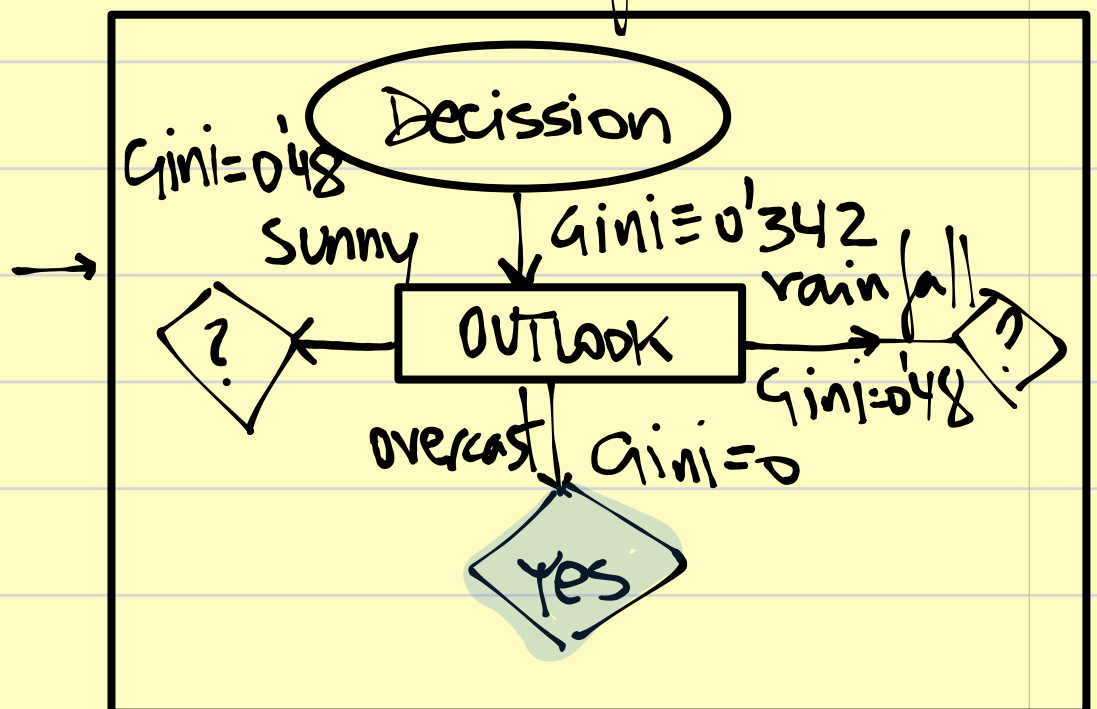
WIND	YES	NO	#
weak	3	3	6
strong	6	2	8

$$Gini(Wind) = \frac{6}{14} \cdot \left[1 - \left[\left(\frac{3}{6} \right)^2 + \left(\frac{3}{6} \right)^2 \right] \right] + \frac{8}{14} \cdot \left[1 - \left[\left(\frac{6}{8} \right)^2 + \left(\frac{2}{8} \right)^2 \right] \right] =$$

$$= 0.428$$

The variable with the lowest Gini is the first node.

	Gini
OUTLOOK	0.342
TEMP	0.439
HUM	0.367
WIND	0.428



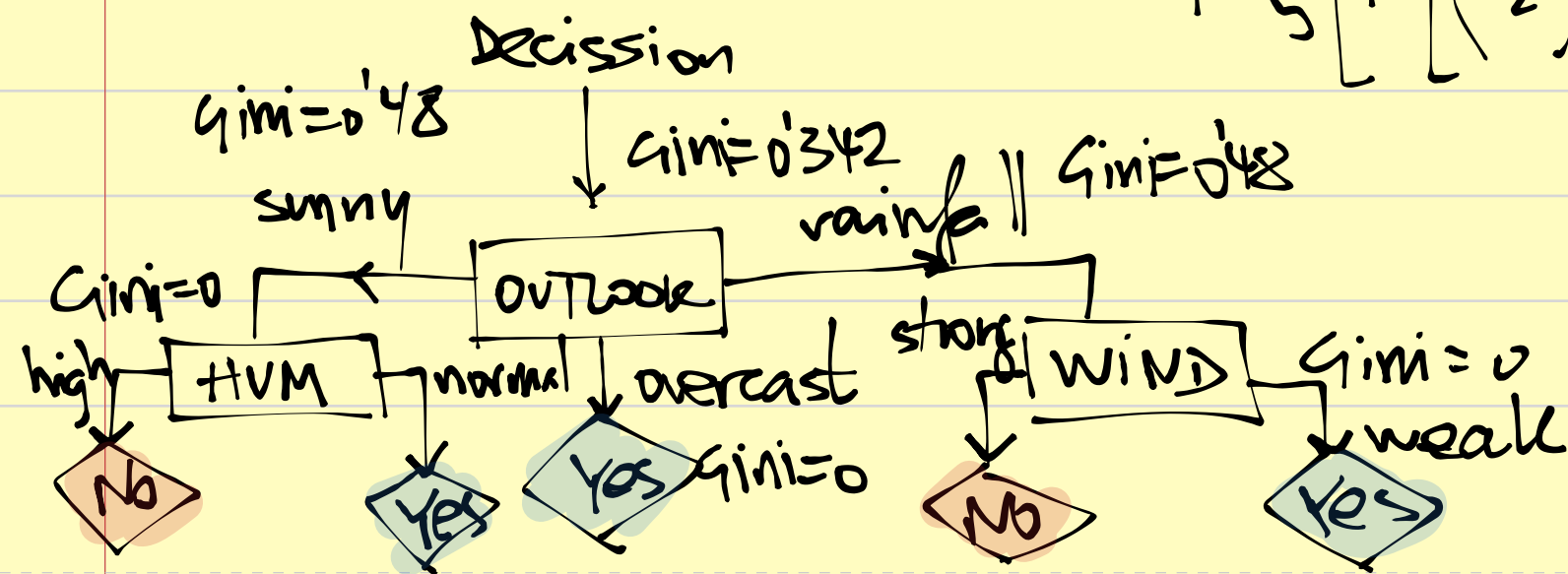
Step 2. Repeat from node 1.

1. outlook (sunny) + Temp	Yes	No	#
hot	0	2	2
mild	1	1	2
cool	1	0	1
			$\frac{5}{5}$

$$\begin{aligned} \text{Gini}(\text{outlook Sunny} + \text{Temp}) &= \frac{2}{5} \left[1 - \left[\left(\frac{0}{2} \right)^2 + \left(\frac{2}{2} \right)^2 \right] \right]_{\text{hot}} + \\ &+ \frac{2}{5} \left[1 - \left[\left(\frac{1}{2} \right)^2 + \left(\frac{1}{2} \right)^2 \right] \right]_{\text{mild}} + \\ &+ \frac{2}{5} \left[1 - \left[\left(\frac{1}{2} \right)^2 + \left(\frac{1}{2} \right)^2 \right] \right]_{\text{cool}} = 0.2 \end{aligned}$$

2. outlook Sunny + Hum	Yes	No	#
high	0	3	3
normal	2	0	2
			$\frac{5}{5}$

$$\begin{aligned} \text{Gini}(\text{outlook Sunny} + \text{Hum}) &= \frac{3}{5} \left[1 - \left[\left(\frac{0}{3} \right)^2 + \left(\frac{3}{3} \right)^2 \right] \right]_{\text{high}} + \\ &+ \frac{2}{5} \left[1 - \left[\left(\frac{2}{2} \right)^2 + \left(\frac{0}{2} \right)^2 \right] \right]_{\text{normal}} = 0 \end{aligned}$$



Step 3. OUTLOOK rainfall + WIND	Yes	No	#
weak	3	0	3
strong	0	2	2
			<u>5</u>

$Gini(Outlook Rain+Wind) = 0$

Example #2. SEX Yes/No?

Exercise #2. SEX Yes/No

	Kitchen clean	Groceries	Strenuous Work	Time Together	D
1.	dirty	big	strong	long	Y
2.	very dirty	small	mild	short	N
3.	dirty	small	mild	short	Y
4.	clean	big	weak	long	Y
5.	very clean	small	strong	long	N
6.	dirty	small	weak	short	N
7.	dirty	big	strong	long	Y
8.	clean	small	weak	short	N

Step 1. First node.

$$Gini(KITCHEN) = \frac{4}{8} \left[1 - \left[\left(\frac{3}{4} \right)^2 + \left(\frac{1}{4} \right)^2 \right] \right] + \frac{1}{8} \left[1 - \left[\left(\frac{1}{1} \right)^2 \right] \right] + \frac{2}{8} \left[1 - \left[\left(\frac{1}{2} \right)^2 + \left(\frac{1}{2} \right)^2 \right] \right] + \frac{1}{8} \left[1 - \left[\left(\frac{1}{1} \right)^2 \right] \right] = 0.3125$$

$$Gini(GROCERIES) = \frac{3}{8} \left[1 - \left[\left(\frac{3}{3} \right)^2 \right] \right] + \frac{5}{8} \left[1 - \left[\left(\frac{4}{5} \right)^2 + \left(\frac{1}{5} \right)^2 \right] \right] = 0.2$$

$$\text{Gini}(\text{Groceries} | \text{Kitchen}) = \frac{2}{5} \cdot 0.5 = 0.2$$

Groceries Small + Time
short
long

Yes	No	4
1	3	4
0	1	1/5

$$\begin{aligned} \text{Gini}(\text{Groceries} | \text{Time}) &= \\ &= \frac{4}{5} \left[1 - \left[\left(\frac{1}{4} \right)^2 + \left(\frac{3}{4} \right)^2 \right] \right] + \frac{1}{5} \cdot 0 = 0.3 \end{aligned}$$

